INNOMATICS®
RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

# PROJECT ON

# Grocery Store Management


**Presented by:**

**SATISH TALLURI**

**Chrishmen .ys**

**Vikram**

**sridhar**

# Introduction

- The Retail and Grocery domain requires effective management of inventory, suppliers, customer orders, employees, and product categories.
- Efficient data handling helps businesses track sales and revenue, monitor product availability, and analyze operations.
- This project simulates a mini grocery store database with interconnected entities like products, orders, and suppliers.
- It offers hands-on experience in understanding data relationships and operational workflows.
- Students will use SQL to extract and transform data, gaining valuable business insights.
- The goal is to improve decision-making and operational efficiency through data analysis.

INNOMATICS® RESEARCH LABS

# Tables

## Suppliers Table

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| sup_id | tinyint | NO | PRI | NULL | |
| sup_name | varchar(255) | YES | | NULL | |
| address | text | YES | | NULL | |

## Employees Table

| Field | Type | Null | Key | Default |
|---|---|---|---|---|
| emp_id | tinyint | NO | PRI | NULL |
| emp_name | varchar(255) | YES | | NULL |
| hire_date | varchar(255) | YES | | NULL |

## Customers Table

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| cust_id | smallint | NO | PRI | NULL | |
| cust_name | varchar(255) | YES | | NULL | |
| address | text | YES | | NULL | |

## Categories Table

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| cat_id | tinyint | NO | PRI | NULL | |
| cat_name | varchar(255) | YES | | NULL | |

INNOMATICS® RESEARCH LABS

# Tables

## Products Table

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| prod_id | tinyint | NO | PRI | NULL | |
| prod_name | varchar(255) | YES | | NULL | |
| sup_id | tinyint | YES | MUL | NULL | |
| cat_id | tinyint | YES | MUL | NULL | |
| price | decimal(10,2) | YES | | NULL | |

## Order details Table

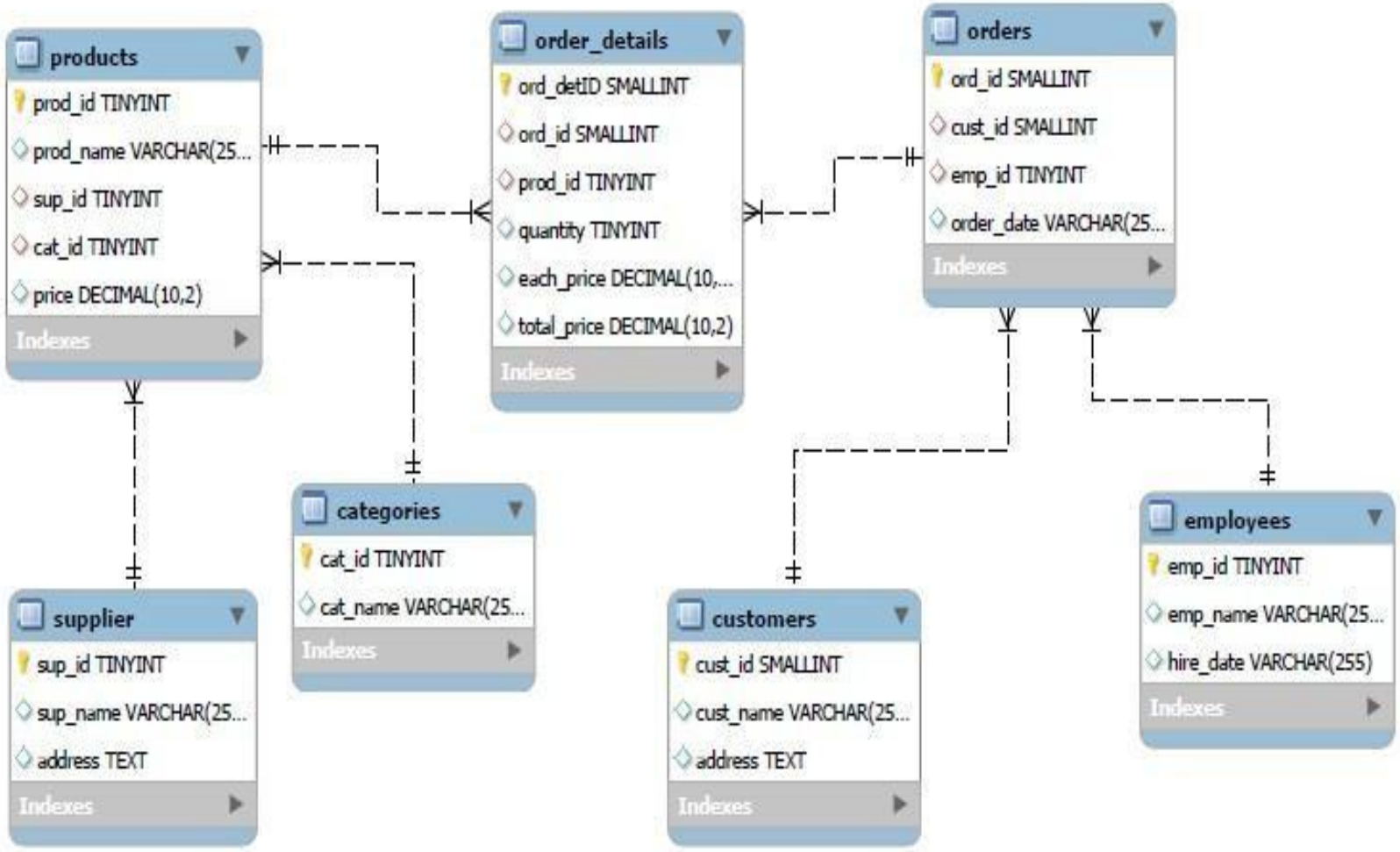
| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| ord_detID | smallint | NO | PRI | NULL | auto_increment |
| ord_id | smallint | YES | MUL | NULL | |
| prod_id | tinyint | YES | MUL | NULL | |
| quantity | tinyint | YES | | NULL | |
| each_price | decimal(10,2) | YES | | NULL | |
| total_price | decimal(10,2) | YES | | NULL | |

## Orders Table

| Field | Type | Null | Key | Default |
|---|---|---|---|---|
| ord_id | smallint | NO | PRI | NULL |
| cust_id | smallint | YES | MUL | NULL |
| emp_id | tinyint | YES | MUL | NULL |
| order_date | varchar(255) | YES | | NULL |

INNOMATICS®
RESEARCH LABS

# ER Diagram



INNOMATICS®
RESEARCH LABS

# Problem Statement

In the dynamic environment of a retail grocery store, managing and analyzing large volumes of data related to products, suppliers, customers, and orders is crucial for efficient operations and informed decision-making. This project aims to design and implement a relational database system that captures essential store activities and enables users to retrieve and analyze data through SQL. The system will support complex queries to identify key business insights such as top-performing products, high-value customers, and revenue patterns, while reinforcing practical SQL skills including joins, aggregations, subqueries, and filtering.

INNOMATICS®
RESEARCH LABS

# 1.How many unique customers have placed orders?

```sql
SELECT
    COUNT(DISTINCT cust_id) AS unique_customers
FROM
    orders;
```

| unique_customers |
| --- |
| 156 |

INNOMATICS®
RESEARCH LABS

# 2.Which customers have placed the highest number of orders?

```sql
SELECT
    c.cust_id, c.cust_name, COUNT(o.ord_id) AS total_orders
FROM
    customers c
        JOIN
    orders o ON c.cust_id = o.cust_id
GROUP BY c.cust_id
ORDER BY total_orders DESC
LIMIT 1;
```

| cust_id | cust_name | total_orders |
|---|---|---|
| 165 | Jyotika | 7 |

INNOMATICS®
RESEARCH LABS

# 3.What is the total and average purchase value per customer?

```sql
SELECT
    o.cust_id,
    c.cust_name,
    SUM(od.total_price) AS total_purchase,
    AVG(od.total_price) AS avg_purchase
FROM
    orders o
        JOIN
    order_details od ON o.ord_id = od.ord_id
        JOIN
    customers c ON o.cust_id = c.cust_id
GROUP BY o.cust_id;
```

| cust_id | cust_name | total_purchase | avg_purchase |
|---|---|---|---|
| 158 | Eshwar Menon | 3061.90 | 765.475000 |
| 129 | Kiran Pillai | 2625.93 | 656.482500 |
| 27 | Chetan Gowda | 5750.59 | 821.512857 |
| 122 | Chetan Reddy | 3869.54 | 1289.846667 |
| 168 | Kasturi | 3865.31 | 1288.436667 |
| 157 | Deepa Gowda | 1745.11 | 872.555000 |
| 125 | Gita Nair | 6305.09 | 1261.018000 |
| 167 | Karishma | 5426.90 | 493.354545 |
| 166 | Kapila | 11099.51 | 1109.951000 |
| 120 | Kiran Iyer | 5588.91 | 698.613750 |
| 145 | Chetan Rao | 5351.18 | 668.897500 |
| 141 | Hari Nair | 5722.71 | 817.530000 |
| 182 | Nikita | 5595.89 | 932.648333 |
| 163 | Esha | 2515.25 | 1257.625000 |
| 94 | Gita Menon | 6581.93 | 731.325556 |
| 21 | Bala Menon | 380.72 | 190.360000 |
| 113 | Chetan Nair | 4044.01 | 577.715714 |
| 56 | Eshwar Rao | 5726.35 | 954.391667 |
| 185 | Girish Gupta | 4778.52 | 1194.630000 |
| 31 | Hari Rao | 6624.78 | 662.478000 |
| 106 | Gita Menon | 5504.08 | 1100.816000 |
| 8 | Deepa Reddy | 7929.13 | 881.014444 |

INNOMATICS® RESEARCH LABS

# 4.Who are the top 5 customers by total purchase amount?

```sql
SELECT
    o.cust_id,
    c.cust_name,
    SUM(od.total_price) AS total_purchase
FROM
    orders o
        JOIN
    order_details od ON o.ord_id = od.ord_id
        JOIN
    customers c ON o.cust_id = c.cust_id
GROUP BY o.cust_id
ORDER BY total_purchase DESC
LIMIT 5;
```

| cust_id | cust_name | total_purchase |
|---|---|---|
| 19 | Chetan Naidu | 11256.82 |
| 166 | Kapila | 11099.51 |
| 67 | Eshwar Rao | 10819.96 |
| 61 | Aditi Rao | 10230.64 |
| 7 | Eshwar Iyer | 9188.45 |

INNOMATICS®
RESEARCH LABS

# 5.How many products exist in each category?

```sql
SELECT
    c.cat_id, c.cat_name, COUNT(p.prod_id) AS total_products
FROM
    categories c
        LEFT JOIN
    products p ON c.cat_id = p.cat_id
GROUP BY c.cat_id;
```

| cat_id | cat_name | total_products |
|---|---|---|
| 1 | Grains & Cereals | 18 |
| 2 | Dairy Products | 6 |
| 3 | Beverages | 17 |
| 4 | Personal Care | 6 |
| 5 | Snacks & Confectioneries | 3 |

INNOMATICS®
RESEARCH LABS

# 6.What is the average price of products by category?

```
SELECT
    c.cat_id, c.cat_name, AVG(p.price) AS avg_price
FROM
    categories c
        LEFT JOIN
    products p ON c.cat_id = p.cat_id
GROUP BY c.cat_id;
```

| cat_id | cat_name | avg_price |
|---|---|---|
| 1 | Grains & Cereals | 287.673333 |
| 2 | Dairy Products | 366.943333 |
| 3 | Beverages | 278.892353 |
| 4 | Personal Care | 364.991667 |
| 5 | Snacks & Confectioneries | 363.336667 |

INNOMATICS®
RESEARCH LABS

# 7.Which products have the highest total sales volume (by quantity)?

```
SELECT
    p.prod_name, SUM(quantity) AS high_sales
FROM
    order_details o
        LEFT JOIN
    products p ON o.prod_id = p.prod_id
GROUP BY prod_name
ORDER BY high_sales DESC
LIMIT 1;
```

| prod_name | high_sales |
|---|---|
| Bath Soap | 60 |

INNOMATICS®
RESEARCH LABS

# 8.What is the total revenue generated by each product?

```
SELECT
    SUM(o.total_price) AS total_revenue, p.prod_name
FROM
    order_details o
        LEFT JOIN
    products p ON o.prod_id = p.prod_id
GROUP BY prod_name;
```

| total_revenue | prod_name |
|---|---|
| 5062.45 | Black Pepper |
| 18561.92 | Cashews |
| 5819.23 | Green Tea |
| 5703.43 | Salt |
| 19695.02 | Moong Dal |
| 7918.87 | Dishwashing Soap |
| 14848.72 | Detergent Powder |
| 11283.92 | Tomato Ketchup |
| 8406.10 | Cumin Seeds |
| 14113.00 | Bath Soap |
| 11100.45 | Mayonnaise |
| 6104.70 | Coffee Powder |
| 13161.31 | Ghee |
| 10084.19 | Chapati |
| 8309.02 | Cinnamon Sticks |
| 20995.92 | Biscuits |
| 10949.84 | Conditioner |
| 16776.90 | Toilet Cleaner |
| 9709.00 | Wheat Flour |
| 3110.56 | Soybean Oil |
| 6752.59 | Mango Pickle |
| 6208.20 | Cardamom |

INNOMATICS®
RESEARCH LABS

# 9.How do product sales vary by category and supplier?

```
SELECT
    c.cat_name, s.sup_name, SUM(o.total_price) AS total_sales
FROM
    order_details o
        JOIN
    products p ON o.prod_id = p.prod_id
        JOIN
    categories c ON p.cat_id = c.cat_id
        JOIN
    supplier s ON p.sup_id = s.sup_id
GROUP BY c.cat_name , s.sup_name
ORDER BY c.cat_name , total_sales DESC;
```

| cat_name | sup_name | total_sales |
|---|---|---|
| Dairy Products | Sai | 50740.60 |
| Dairy Products | Aarya | 18519.61 |
| Dairy Products | Karthik | 11100.45 |
| Grains & Cereals | Aarya | 67701.10 |
| Grains & Cereals | Karthik | 39473.49 |
| Grains & Cereals | Suresh | 26248.89 |
| Grains & Cereals | Sai | 18018.02 |
| Grains & Cereals | Aarav Sharma | 6104.70 |
| Household | Karthik | 22767.59 |
| Household | Sai | 16776.90 |
| Personal Care | Aarya | 69378.41 |
| Personal Care | Sai | 10949.84 |
| Personal Care | Suresh | 10132.75 |
| Snacks & Conf... | Aarya | 65538.71 |
| Snacks & Conf... | Suresh | 65307.14 |
| Snacks & Conf... | Aarav Sharma | 26948.15 |
| Snacks & Conf... | Sai | 17103.15 |
| Snacks & Conf... | Karthik | 8520.43 |

INNOMATICS®
RESEARCH LABS

# 10.How many orders have been placed in total?

```sql
SELECT
    COUNT(*) AS total_orders
FROM
    orders;
```

| total_orders |
| --- |
| 300 |

INNOMATICS®
RESEARCH LABS

# 11.What is the average value per order?

```sql
SELECT
    SUM(total_price) / COUNT(DISTINCT ord_id) AS avg_order_value
FROM
    order_details;
```

| avg_order_value |
| --- |
| 2153.632539 |

INNOMATICS®
RESEARCH LABS

# 12.On which dates were the most orders placed?

```sql
SELECT
    order_date, COUNT(ord_id) AS orders_count
FROM
    orders
GROUP BY order_date
ORDER BY orders_count DESC
LIMIT 1;
```

| order_date | orders_count |
|---|---|
| 9/10/2022 | 4 |

INNOMATICS®
RESEARCH LABS

# 13.What are the monthly trends in order volume and revenue?

```sql
SELECT
    LEFT(o.order_date, 7) AS order_month,
    COUNT(DISTINCT o.ord_id) AS order_volume,
    SUM(od.total_price) AS total_revenue
FROM
    orders o
        JOIN
    order_details od ON o.ord_id = od.ord_id
GROUP BY order_month
ORDER BY order_month;
```

| order_month | order_volume | total_revenue |
|---|---|---|
| 1/1/202 | 1 | 711.73 |
| 1/10/20 | 1 | 133.51 |
| 1/11/20 | 2 | 2217.16 |
| 1/12/20 | 1 | 3489.70 |
| 1/14/20 | 2 | 3333.21 |
| 1/15/20 | 1 | 5252.28 |
| 1/16/20 | 3 | 6340.93 |
| 1/18/20 | 1 | 1449.57 |
| 1/22/20 | 1 | 2168.52 |
| 1/23/20 | 2 | 3957.10 |
| 1/24/20 | 2 | 7187.71 |
| 1/25/20 | 1 | 3485.80 |
| 1/28/20 | 3 | 4703.02 |
| 1/30/20 | 2 | 7433.69 |
| 1/31/20 | 1 | 3125.69 |
| 1/5/202 | 2 | 5466.54 |
| 1/6/202 | 1 | 1944.07 |
| 1/7/202 | 2 | 5553.92 |
| 1/8/202 | 1 | 2358.30 |
| 10/10/2 | 2 | 2378.37 |
| 10/14/2 | 1 | 1667.05 |
| 10/19/2 | 2 | 7406.59 |

INNOMATICS®
RESEARCH LABS

# 14.How do order patterns vary across weekdays and months?

```sql
UPDATE orders
SET order_date = DATE_FORMAT(STR_TO_DATE(order_date, '%m/%d/%Y'), '%Y-%m-%d');

SELECT
    DAYNAME(STR_TO_DATE(order_date, '%Y-%m-%d')) AS weekday,
    MONTHNAME(STR_TO_DATE(order_date, '%Y-%m-%d')) AS month,
    COUNT(ord_id) AS total_orders
FROM
    orders
GROUP BY weekday , month
ORDER BY weekday , month;
```

| weekday | month | total_orders |
|---|---|---|
| Friday | April | 4 |
| Friday | August | 2 |
| Friday | December | 5 |
| Friday | February | 2 |
| Friday | January | 8 |
| Friday | July | 5 |
| Friday | June | 2 |
| Friday | March | 6 |
| Friday | May | 5 |
| Friday | November | 6 |
| Friday | October | 1 |
| Friday | September | 6 |
| Monday | April | 1 |
| Monday | August | 4 |
| Monday | December | 7 |
| Monday | February | 4 |
| Monday | January | 5 |
| Monday | June | 3 |

INNOMATICS®
RESEARCH LABS

# 15.How many suppliers are there in the database?

```
SELECT
    COUNT(*)
FROM
    supplier;
```

| COUNT(*) |
| --- |
| 5 |

INNOMATICS®
RESEARCH LABS

# 16.Which supplier provides the most products?

```
SELECT
    s.sup_id, s.sup_name, COUNT(p.prod_id) AS product_count
FROM
    supplier s
        JOIN
    products p ON s.sup_id = p.sup_id
GROUP BY s.sup_id
ORDER BY product_count DESC
LIMIT 1;
```

| sup_id | sup_name | product_count |
|---|---|---|
| 3 | Aarya | 18 |

INNOMATICS®
RESEARCH LABS

# 17.What is the average price of products from each supplier?

```
SELECT
    s.sup_id, s.sup_name, AVG(p.price)
FROM
    products p
        JOIN
    supplier s ON s.sup_id = p.sup_id
GROUP BY s.sup_id;
```

| sup_id | sup_name | AVG(p.price) |
|---|---|---|
| 1 | Aarav Sharma | 271.366667 |
| 2 | Sai | 342.672000 |
| 3 | Aarya | 319.326667 |
| 4 | Suresh | 281.818000 |
| 5 | Karthik | 288.225556 |

INNOMATICS®
RESEARCH LABS

# 18.Which suppliers contribute the most to total product sales (by revenue)

```
SELECT
    s.sup_id, s.sup_name, SUM(od.total_price) AS total_revenue
FROM
    supplier s
        JOIN
    products p ON s.sup_id = p.sup_id
        JOIN
    order_details od ON p.prod_id = od.prod_id
GROUP BY s.sup_id
ORDER BY total_revenue DESC
LIMIT 1;
```

| | sup_id | sup_name | total_revenue |
|---|---|---|---|
| ▶ | 3 | Aarya | 221137.83 |

INNOMATICS®
RESEARCH LABS

# 19.How many employees have processed orders?

```
SELECT
    COUNT(DISTINCT emp_id)
FROM
    orders;
```

| | COUNT(DISTINCT emp_id) |
|---|---|
| ▶ | 10 |

INNOMATICS®
RESEARCH LABS

# 20.Which employees have handled the most orders?

```sql
SELECT
    e.emp_id, e.emp_name, COUNT(o.ord_id) AS ordered_most
FROM
    orders o
        JOIN
    employees e ON o.emp_id = e.emp_id
GROUP BY e.emp_id
ORDER BY ordered_most DESC
LIMIT 1;
```

| | emp_id | emp_name | ordered_most |
|---|---|---|---|
| ▶ | 8 | Diya Sharma 1 | 38 |

INNOMATICS® RESEARCH LABS

# 21.What is the total sales value processed by each employee?

```sql
SELECT
    e.emp_id, e.emp_name, SUM(od.total_price) AS total_sales
FROM
    employees e
        JOIN
    orders o ON e.emp_id = o.emp_id
        JOIN
    order_details od ON o.ord_id = od.ord_id
GROUP BY e.emp_id
ORDER BY total_sales DESC;
```

| emp_id | emp_name | total_sales |
|---|---|---|
| 2 | Aditya Singh 1 | 79252.29 |
| 6 | Zara Verma 1 | 71562.76 |
| 8 | Diya Sharma 1 | 67241.85 |
| 3 | Pari Kumar 1 | 66818.39 |
| 9 | Arjun Kumar 1 | 54018.31 |
| 1 | Aarav Kumar 1 | 52602.88 |

INNOMATICS® RESEARCH LABS

# 22.What is the average order value handled per employee?

```sql
SELECT
    e.emp_id,
    e.emp_name,
    SUM(od.total_price) / COUNT(DISTINCT o.ord_id) AS avg_value
FROM
    employees e
        JOIN
    orders o ON e.emp_id = o.emp_id
        JOIN
    order_details od ON o.ord_id = od.ord_id
GROUP BY e.emp_id
ORDER BY avg_value DESC;
```

| emp_id | emp_name | avg_value |
|---|---|---|
| 1 | Aarav Kumar 1 | 2768.572632 |
| 6 | Zara Verma 1 | 2650.472593 |
| 2 | Aditya Singh 1 | 2330.949706 |
| 3 | Pari Kumar 1 | 2227.279667 |
| 7 | Vihaan Singh 1 | 2112.081739 |
| 9 | Arjun Kumar 1 | 2077.627308 |
| 8 | Diya Sharma 1 | 2037.631818 |
| 10 | Arjun Verma 1 | 1835.842000 |
| 5 | Pari Sharma 1 | 1833.373636 |
| 4 | Aditya Verma 1 | 1554.750455 |

INNOMATICS® RESEARCH LABS

# 23.What is the relationship between quantity ordered and total price?

```sql
SELECT
  quantity,
  AVG(od.total_price) AS avg_total_price,
  COUNT(*) AS order_count
FROM order_details od
GROUP BY quantity
ORDER BY quantity;
```

| quantity | avg_total_price | order_count |
|---|---|---|
| 1 | 319.274516 | 124 |
| 2 | 595.716667 | 117 |
| 3 | 898.266377 | 138 |
| 4 | 1338.607143 | 105 |
| 5 | 1530.401121 | 116 |

INNOMATICS®
RESEARCH LABS

# 24.What is the average quantity ordered per product?

```sql
SELECT
    p.prod_id,
    p.prod_name,
    COUNT(od.ord_id) AS total_orders,
    AVG(od.quantity) AS avg_quantity_ordered
FROM
    products p
        JOIN
    order_details od ON p.prod_id = od.prod_id
GROUP BY p.prod_id
ORDER BY avg_quantity_ordered DESC;
```

| prod_id | prod_name | total_orders | avg_quantity_ordered |
|---|---|---|---|
| 40 | Butter | 9 | 4.5556 |
| 31 | Toothpaste | 12 | 3.6667 |
| 46 | Potato Chips | 15 | 3.6000 |
| 42 | Tomato Ketchup | 10 | 3.5000 |
| 22 | Mustard Seeds | 13 | 3.4615 |
| 3 | Moong Dal | 15 | 3.4000 |
| 6 | Ghee | 8 | 3.3750 |
| 9 | Mango Pickle | 11 | 3.3636 |
| 32 | Bath Soap | 18 | 3.3333 |
| 45 | Chili Sauce | 8 | 3.2500 |
| 18 | Salt | 8 | 3.2500 |
| 34 | Facial Tissue | 14 | 3.2143 |
| 1 | Basmati Rice | 10 | 3.2000 |
| 26 | Detergent Pow... | 10 | 3.2000 |
| 35 | Mouth wash | 10 | 3.2000 |
| 16 | Sugar | 5 | 3.2000 |
| 13 | Green Tea | 11 | 3.1818 |
| 47 | Chocolate Bar | 11 | 3.1818 |
| 49 | Instant Noodles | 11 | 3.1818 |

INNOMATICS®
RESEARCH LABS

# 25.How does the unit price vary across products and orders?

```
SELECT
    p.prod_id,
    p.prod_name,
    od.each_price AS unit_price,
    COUNT(od.ord_id) AS times_ordered
FROM
    products p
        JOIN
    order_details od ON p.prod_id = od.prod_id
GROUP BY p.prod_id , od.each_price
ORDER BY p.prod_id , od.each_price;
```

| prod_id | prod_name | unit_price | times_ordered |
|---|---|---|---|
| 1 | Basmati Rice | 358.98 | 10 |
| 2 | Wheat Flour | 255.50 | 15 |
| 3 | Moong Dal | 386.18 | 15 |
| 4 | Chickpeas | 353.50 | 7 |
| 5 | Soybean Oil | 172.81 | 11 |
| 6 | Ghee | 487.46 | 8 |
| 7 | Paneer | 484.27 | 11 |
| 8 | Yogurt | 111.61 | 7 |
| 9 | Mango Pickle | 182.50 | 11 |
| 10 | Mixed Vegetable Pickle | 133.51 | 7 |
| 11 | Almonds | 315.57 | 9 |
| 12 | Cashews | 441.95 | 16 |
| 13 | Green Tea | 166.26 | 11 |
| 14 | Masala Tea | 380.85 | 10 |
| 15 | Coffee Powder | 179.55 | 14 |
| 16 | Sugar | 409.62 | 5 |
| 17 | Jaggery | 200.85 | 12 |

INNOMATICS®
RESEARCH LABS

# conclusion

The Grocery Store Management System using SQL helps in organizing and managing grocery store operations efficiently. It reduces manual work and improves accuracy in handling product, customer, and sales data. By using SQL queries and relational database concepts, the system allows quick data retrieval, better inventory control, and effective sales management. Overall, it makes grocery store operations faster, more reliable, and easier to manage.

INNOMATICS®
RESEARCH LABS

# THANK YOU